Студент) групи КН-115

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1. 10)

2. 15)

3. 8)

4. 5)

5. 6)

6. 7)

7. 8)

8. 15)

9. 10)

10. 3)

11. 4)

1.10) $\begin{cases} -2x_1 + x_2 - 3x_3 = 4 & (1) \\ x_1 - x_2 + 2x_3 = 3 & (2) \\ x_1 + x_3 = -7 & (3) \\ -3x_1 + 2x_2 - 5x_3 = 1 & (4) \end{cases}$

(1) - (2) = (4)

$2x_1 - x_2 + 3x_3 = -4$

$-x_1 + x_2 - 2x_3 = -3$ $\Big| + \Rightarrow$ $x_1 + x_3 = -7$ $\Big| \Rightarrow$ $-(1) - (2) = (3)$

Лише 2 незалежних рівняння

$\begin{cases} x_1 - x_2 + 2x_3 = 3 \\ x_1 + x_3 = -7 \end{cases}$ Нехай $x_1 = a \Rightarrow$

$\Rightarrow x_3 = -7 - a$

$x_2 = a + 2(-7-a) - 3$

$x_2 = a - 14 - 2a - 3 = -a - 17$

В-дь: $\begin{cases} x_1 = a \\ x_2 = -7 - a \\ x_3 = -a - 17 \end{cases}$

2.15) $\vec{a}(2;-2;3)$ $\quad \vec{a} \perp \vec{d}: \; 2x-2y+3z=0$

$\vec{b}(2;1;2)$ $\quad \vec{b} \perp \vec{d}: \; 2x+y+2z=0$

$\vec{c}(3;-1;4)$ $\quad \vec{d}\vec{c}=-4: \; 3x-y+4z=-4$

$\vec{d}(x;y;z)$

$$\begin{cases} 2x-2y+3z=0 \\ 2x+y+2z=0 \\ 3x-y+4z=-4 \end{cases} \Rightarrow \begin{array}{l} 2x-2y+3z-2x-y-2z=0 \\ -3y+z=0 \end{array}$$

$$3y=z$$

$$\begin{cases} 2x+y+6y=0 \\ 3x-y+12y=-4 \end{cases} \Rightarrow \begin{cases} 2x+7y=0 \\ 3x+11y=-4 \end{cases} \Rightarrow y=\frac{-2x}{7}$$

$$3x-\frac{22x}{7}=-4 \Rightarrow \frac{21x-22x}{7}=-4$$

$$\frac{-x}{7}=-4 \Rightarrow x=28 \Rightarrow y=-2\cdot 4=-8$$

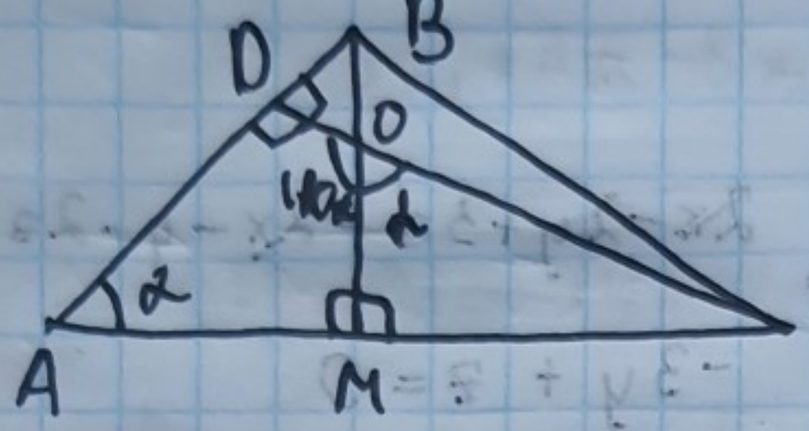
$$z=3y=-24$$

B-k: $\vec{d}(28;-8;-24)$

3.8) $A(3;1;1)$ $\quad \angle(\overline{CD};BM) = \varphi = ?$

$B(4;-1;2)$

$C(-3;2;-1)$



ADOM: $90 + \alpha + 90 + \beta = 360$

$\beta = 180 - \alpha \Rightarrow$

$\Rightarrow \varphi = \alpha = \angle PAM$

$\overline{AB}\ (1;-2;1) \quad |\overline{AB}| = \sqrt{4+1+1} = \sqrt{6}$

$\overline{AC}\ (-6;1;-2) \quad |\overline{AC}| = \sqrt{36+1+4} = \sqrt{41}$

$\overline{AB} \cdot \overline{AC} = -6 - 2 \cdot 2 = -10$

$\cos\alpha = \dfrac{\overline{AB} \cdot \overline{AC}}{|\overline{AB}|\,|\overline{AC}|} = \dfrac{-10}{\sqrt{41} \cdot \sqrt{6}} = \dfrac{-10}{\sqrt{246}}$

$\alpha = \arccos\left(\dfrac{-10}{\sqrt{246}}\right) > 90° \Rightarrow \quad \varphi = 180° - \arccos\left(\dfrac{-10}{\sqrt{246}}\right)$

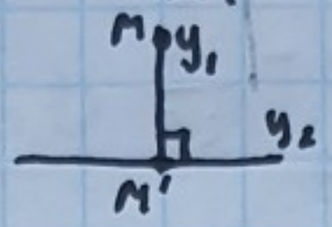
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B-go: $\angle(CD;BM) = 180° - \arccos\left(\dfrac{-10}{\sqrt{246}}\right)$

45) $M(-3; 8)$ $\quad 3x+2y+6=0$

$y = \frac{-3}{2}x - 3$ $\quad K_1 = -\frac{1}{K_2} \Rightarrow K_2 = \frac{2}{3}$

$y = \frac{2}{3}x + c \Rightarrow 8 = -2 + c \Rightarrow c = 10$



$$\begin{cases} y_1 = -\frac{3}{2}x - 3 \\ y_2 = \frac{2}{3}x + 10 \end{cases} \Rightarrow$$ перетин прямих є тч та у т M' – т. проекції.

$\frac{2}{3}x + 10 = \frac{-3x}{2} - 3 \qquad \frac{2x+30}{3} = -\frac{3x-6}{2}$

$4x + 60 = -9x - 18 \Rightarrow 13x = -78 \quad x = -6$

$y = -\frac{2 \cdot 6}{3} + 10 = -4 + 10 = 6$

В-дь: $M'(-6; 6)$

5.6) $M_1(1; -2; 1)$

$M_2(1; -2; 3)$

$\vec{a}(-1; 3; -2)$

$M(x; y; z)$

$\Rightarrow \overrightarrow{M_1M}\ (x-1; y+2; z-1)$

$\overrightarrow{M_1M_2}\ (0; 0; 2)$

$$(\overrightarrow{M_1M}; \overrightarrow{M_1M_2}; \vec{a}) = \begin{vmatrix} x-1 & y+2 & z-1 \\ 0 & 0 & 2 \\ -1 & 3 & -2 \end{vmatrix} = 0$$

$0 + 0 - 2y - 4 - 0 - 0 - 6x + 6 = 0$

$-6x - 2y + 2 = 0$

$6x + 2y - 2 = 0$

$3x + y - 1 = 0$

B-gb: $3x + y - 1 = 0$

6. 7) $4x^2+4xy+y^2+6x+4y+2=0$

$$A=\begin{pmatrix}4 & 2\\ 2 & 1\end{pmatrix} \qquad \begin{vmatrix}4-\lambda & 2\\ 2 & 1-\lambda\end{vmatrix}=0$$

$$(4-\lambda)(1-\lambda)-4=0$$

$\lambda=0$ $\qquad$ $\lambda=5$

$\lambda_1=0$:

$$\begin{cases}4x_1+2x_2=0\\ 2x_1+x_2=0\end{cases}$$

$2x_1=-x_2$

Нехай $x_1=1 \Rightarrow x_2=-2$

$a\ (1;-2)$

$|a|=\sqrt{5}$

$\vec{a_0}\ \left(\frac{1}{\sqrt{5}};-\frac{2}{\sqrt{5}}\right)$

$\lambda_2=5$

$$\begin{cases}-x_1+2x_2=0\\ 2x_1-4x_2=0\end{cases}$$

$x_1=2x_2$

Нехай $x_1=2 \Rightarrow x_2=1$

$b\ (2;1)$

$|b|=\sqrt{5}$

$\vec{b_0}\ \left(\frac{2}{\sqrt{5}};\frac{1}{\sqrt{5}}\right)$

$$B=\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}$$

$$\begin{cases}x=\frac{1}{\sqrt{5}}x'+\frac{2}{\sqrt{5}}y'\\ y=\frac{-2}{\sqrt{5}}x'+\frac{1}{\sqrt{5}}y'\end{cases}$$

$$Q=\begin{pmatrix}0 & 0\\ 0 & 5\end{pmatrix} \Rightarrow f(x;y)=5y'^2$$

$5y'^2+6x+4y+2=0$

$5y'^2+\frac{6}{\sqrt{5}}x'+\frac{12}{\sqrt{5}}y'-\frac{8}{\sqrt{5}}x'+\frac{4}{\sqrt{5}}y'+2=0$

$$5y'^2 - \frac{2}{\sqrt{5}}x' + \frac{16y'}{\sqrt{5}} + 2 = 0 \qquad 2 \cdot \frac{\sqrt{5}}{5}$$

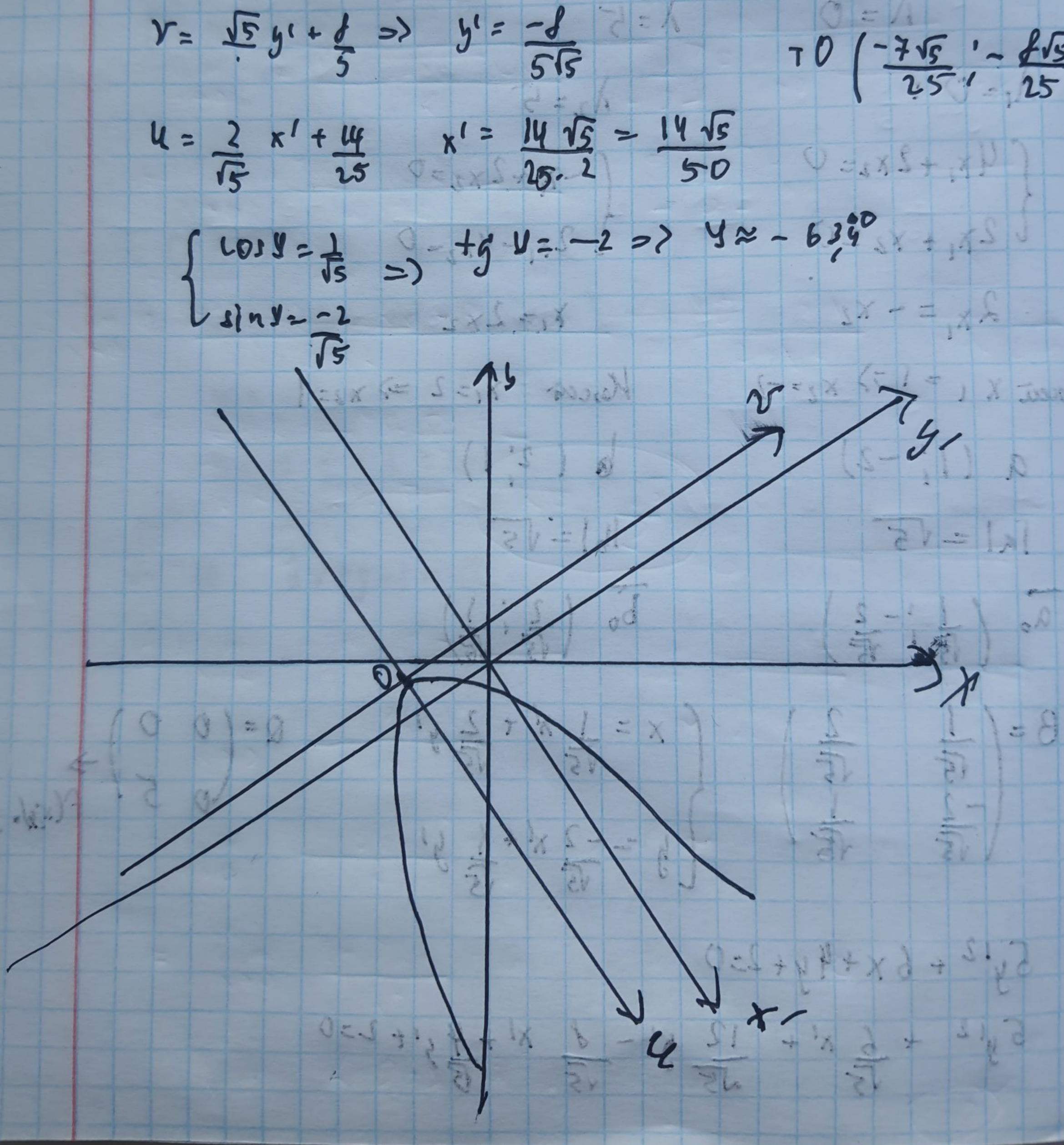
$$\left(\sqrt{5}y' + \frac{8}{5}\right)^2 - \frac{64}{25} + 2 - \frac{2}{\sqrt{5}}x' = 0$$

$$\left(\sqrt{5}y' + \frac{8}{5}\right)^2 = \frac{2}{\sqrt{5}}x' + \frac{14}{25}$$

$$v = \sqrt{5}y' + \frac{8}{5} \Rightarrow y' = \frac{-8}{5\sqrt{5}} \qquad T\,O\left(\frac{-7\sqrt{5}}{25}; -\frac{8\sqrt{5}}{25}\right)$$

$$u = \frac{2}{\sqrt{5}}x' + \frac{14}{25} \qquad x' = \frac{14\sqrt{5}}{25 \cdot 2} = \frac{14\sqrt{5}}{50}$$

$$\begin{cases} \cos\gamma = \frac{1}{\sqrt{5}} \\ \sin\gamma = \frac{-2}{\sqrt{5}} \end{cases} \Rightarrow tg\,\gamma = -2 \Rightarrow \gamma \approx -63{,}4^0$$



7.8) $\lim\limits_{n\to\infty}(5^{-n}+1)=1$

$\lim\limits_{x\to\infty} x_n = a \Rightarrow \forall \varepsilon>0\ \exists\, n_\varepsilon \in N:\ \forall n>n_\varepsilon\ |x_n-a|<\varepsilon$

Задаємо довільно мале $\varepsilon<0$; визначимо n_ε: $|x_n-a|<\varepsilon$

$$\left|\frac{1}{5^n}+1-1\right|<\varepsilon \Rightarrow \frac{1}{5^n}<\varepsilon \Rightarrow 5^n>\frac{1}{\varepsilon}$$

$$\log_5 5^n > \log_5 \frac{1}{\varepsilon} \Rightarrow n > \log_5 \frac{1}{\varepsilon}$$

$n_\varepsilon = \left[\log_5 \frac{1}{\varepsilon}\right]$ = найб. ціле число, що не перевищує дріб $\Rightarrow$

$\Rightarrow \forall n>n_\varepsilon \left|\frac{1}{5^n}+1-1\right|<\varepsilon \Rightarrow \lim\limits_{n\to\infty} 5^{-n}+1=1$

8.15) а) $(\cos 2x - \cos 3x)^3 \sim kx^6, \quad x \to 0$

$$\lim_{x\to 0} \frac{(\cos 2x - \cos 3x)^3}{kx^6} = \left[\frac{0}{0}\right] = \lim_{x\to 0} \frac{3(\cos 2x - \cos 3x)^2 \cdot 2(-2\sin 2x + 3\sin 3x)}{6x^5 k}$$

$$= \lim_{x\to 0} \frac{3(\cos 2x - \cos 3x)^2 (9x - 4x)}{6x^5 k} =$$

$$= \lim_{x\to 0} \frac{15(\cos 2x - \cos 3x)^2}{6x^4 k} = \left[\frac{0}{0}\right] = \text{аналогічно} \lim_{x\to 0} \frac{150(\cos 2x - \cos 3x)}{24x^{\cdot 2} k} =$$

$$= \lim_{x\to 0} \frac{75(\cos 2x - \cos 3x)}{12x^2 k} = \left[\frac{0}{0}\right] =$$

$$= \lim_{x\to 0} \frac{75(-2\sin 2x + 3\sin 3x)}{24xk} = \lim_{x\to 0} \frac{75(5x}{24xk} = \lim_{x\to 0} \frac{75 \cdot 5}{24k} = 1$$

$$24k = 75 \cdot 5$$

$$8k = 25 \cdot 5 \qquad k = \frac{125}{8}$$

б) $(1 + \operatorname{tg}^2 6x)^2 = O(x^k)$

$$\lim_{x\to 0} \frac{(1 + \operatorname{tg}^2 6x)^2}{x^k} = \left| \operatorname{tg}^2 6x \sim 36x^2 \right| =$$

$$= \lim_{x\to 0} \frac{(1 + 36x^2)^2}{x^k} = \left| (1 + 36x^2)^2 \sim 1 + 72x^2 \right| =$$

$$= \lim_{x\to 0} \frac{1 + 72x^2}{x^k} = a \qquad a \neq 0;\ a \neq \infty \quad a - \text{число}$$

Коли $k = 0 \Rightarrow a = 1$ – еквівал. – інших таких умов а немає

$k = 0$

9.101 а) $y = x^3 \cos\frac{1}{x}$ $\quad x \neq 0$

$\lim\limits_{x\to 0-0} x^3 \cos\frac{1}{x} = \lim\limits_{x\to 0+0} x^3 \cos\frac{1}{x} =$ невизначеність $\Rightarrow$

$\Rightarrow$ розрив II роду т. розриву $x = 0$

Неперервна на проміжках: $(-\infty; 0)$ та $(0; +\infty)$

б) $y = \frac{x+3}{x^2-x-6} = \frac{x+3}{(x-3)(x+2)}$ $\quad x \neq 3$, $x \neq -2$

$D = 1 + 24 = 5$

$x = \frac{1 \pm 5}{2} = 3, -2$

$\lim\limits_{x\to 3-0} \frac{x+3}{(x-3)(x+2)} = \frac{6}{0 \cdot 5} = \infty \Rightarrow$ розрив II роду

$\lim\limits_{x\to -2-0} \frac{x+3}{(x-3)(x+2)} = \frac{1}{-5 \cdot 0} = \pm\infty \Rightarrow$ розрив II роду

Т. розриву: $x = 3$ та $x = -2$

Неперервна на проміжках $(-\infty; -2)$ та $(-2; 3)$ та $(3; +\infty)$

10.3)

а) $\lim\limits_{x\to -8} \dfrac{\sqrt{9-2x}-5}{\sqrt[3]{x}+2} = \lim\limits_{x\to -8} \dfrac{(\sqrt{9-2x}-5)(\sqrt{9-2x}+5)}{(\sqrt[3]{x}+2)(\sqrt{9-2x}+5)} =$

$= \lim\limits_{x\to -8} \dfrac{-16-2x}{(\sqrt[3]{x}+2)(\sqrt{9-2x}+5)} = \lim\limits_{x\to -8} \dfrac{-2(x+8)}{(\sqrt[3]{x}+2)(\sqrt{9-2x}+5)} =$

$= \lim\limits_{x\to -8} \dfrac{-2(\sqrt[3]{x^2}-2\sqrt[3]{x}+4)}{\sqrt{9-2x}+5} = \dfrac{-2(4+4+4)}{10} = -\dfrac{12}{5}$

б) $\lim\limits_{x\to 0} \dfrac{(1-\cos x)^3}{2x^2 \sin^4 x} = |\sin x \sim x| = \lim\limits_{x\to 0} \dfrac{(1-\cos x)^3}{2x^6} =$

$= \lim\limits_{x\to 0} \dfrac{\left(2\sin^2\frac{x}{2}\right)^3}{2x^6} = \lim\limits_{x\to 0} \dfrac{8\sin^6\frac{x}{2}}{2x^6} = \lim\limits_{x\to 0} \dfrac{\frac{8}{2^6}x^6}{2x^6} =$

$= \dfrac{8}{27} = \dfrac{2^3}{27} = \dfrac{1}{2^4} = \dfrac{1}{16}$

в) $\lim\limits_{x\to\infty} \left(\sqrt{x^2+4x}-\sqrt{x^2-2x}\right) = \lim\limits_{x\to\infty} \dfrac{x^2-x^2+2x+2x}{\sqrt{x^2+2x}+\sqrt{x^2-2x}} =$

$= \lim\limits_{x\to\infty} \dfrac{4x}{\sqrt{x(x+2)}+\sqrt{x^2-2x}} = \lim\limits_{x\to\infty} \dfrac{4}{\sqrt{1+\frac{2}{x}}+\sqrt{1-\frac{2}{x}}} = \dfrac{4}{2} = 2$

11. 4) а) $y = 5^{\arccos 2x^3}$

$$y' = \ln 5 \cdot 5^{\arccos 2x^3} \cdot \frac{-1}{\sqrt{1-4x^6}} \cdot 6x^2 =$$

$$y' = -\frac{6x^2 \ln 5 \cdot 5^{\arccos 2x^3}}{\sqrt{1-4x^6}}$$

б) $y = \sin 2x^{\frac{1}{3x}}$

$$\frac{y'}{y} = \left(\frac{\ln(\sin 2x)}{3x} \cdot \frac{1}{3x}\right)'$$

$$y' = \sin 2x^{\frac{1}{3x}} \left(\frac{1}{\sin 2x} \cdot \cos 2x \cdot 2 \cdot \frac{1}{3x} - \frac{\ln \sin 2x}{3x^2}\right)$$

$$y' = \sin 2x^{\frac{1}{3x}} \left(\frac{2 \operatorname{ctg} 2x}{3x} - \frac{\ln \sin 2x}{3x^2}\right)$$